

Problem Set 7

Econ 5253 — Spring 2026

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Question 5: Dropping Missing Observations

After loading `wages.csv` (2,246 observations), we drop all rows where either `hgc` or `tenure` is missing (17 observations total), leaving **2,229 observations** for analysis.

Question 6: Summary Statistics and Missing Data Discussion

Table 1 presents summary statistics for the cleaned sample.

Table 1: Summary Statistics ($N = 2,229$)					
	Count	Mean	Std. Dev.	Min	Max
logwage	1669	1.6252	0.3855	0.0049	2.2615
hgc	2229	13.1014	2.5243	0.0000	18.0000
college	2229	0.2378	0.4258	0	1
tenure	2229	5.9706	5.5072	0.0000	25.9167
age	2229	39.1516	3.0620	34.0000	46.0000
married	2229	0.6420	0.4795	0	1

Missing rate for logwage: Of the 2,229 observations, 560 have missing log wages, for a missing rate of **25.1%**.

MCAR, MAR, or MNAR? The `logwage` variable is most likely **Missing Not at Random (MNAR)**. Women with lower wages—e.g. those in informal or part-time employment—are less likely to have their wages recorded. This means the probability of missingness is correlated with the unobserved value of `logwage` itself, which rules out both MCAR (missingness independent of all data) and MAR (missingness dependent only on observed covariates).

Question 7: Imputation Methods and Regression Results

We estimate the following model under four imputation strategies:

$$\log w_i = \beta_0 + \beta_1 \text{hgc}_i + \beta_2 \text{college}_i + \beta_3 \text{tenure}_i + \beta_4 \text{tenure}_i^2 + \beta_5 \text{age}_i + \beta_6 \text{married}_i + \varepsilon_i \quad (1)$$

The four methods are: (1) **complete cases** (listwise deletion, assumes MCAR); (2) **mean imputation** (replace missing `logwage` with the sample mean); (3) **predicted value imputation** (impute using fitted values from the complete-cases regression, consistent with MAR); and (4) **multiple imputation** (5 imputed datasets with added residual noise, pooled via Rubin's rules).

Table 2 presents all four sets of estimates. The true value of $\beta_1 = 0.093$.

Table 2: Returns to Schooling under Alternative Imputation Methods

	Complete Cases	Mean Imputation	Predicted Imputation	Multiple Imputation
Intercept	0.6567*** (0.1301)	0.8490*** (0.1029)	0.6567*** (0.0991)	0.7089*** (0.1133)
hgc ($\hat{\beta}_1$)	0.0624*** (0.0054)	0.0497*** (0.0043)	0.0624*** (0.0042)	0.0590*** (0.0048)
college	-0.1452*** (0.0345)	-0.1682*** (0.0257)	-0.1452*** (0.0247)	-0.1250*** (0.0283)
tenure	0.0495*** (0.0052)	0.0382*** (0.0039)	0.0495*** (0.0037)	0.0498*** (0.0043)
tenure ²	-0.0016*** (0.0003)	-0.0013*** (0.0002)	-0.0016*** (0.0002)	-0.0016*** (0.0002)
age	0.0004 (0.0027)	0.0002 (0.0022)	0.0004 (0.0021)	0.0002 (0.0024)
married	0.0220 (0.0177)	0.0268** (0.0136)	0.0220* (0.0131)	0.0143 (0.0150)
R^2	0.2084	0.1473	0.2767	0.2158
N	1669	2229	2229	2229

Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Discussion of $\hat{\beta}_1$ across models:

All four estimates of $\hat{\beta}_1$ fall below the true value of 0.093, consistent with MNAR: lower-wage observations are missing disproportionately, compressing the observed variation in log wages and attenuating the estimated returns to schooling.

Complete cases (0.0624): Dropping missing observations yields a $\hat{\beta}_1$ notably below

the true value. Because low-wage workers are more likely to be missing, the MNAR pattern pulls the estimate downward.

Mean imputation (0.0497): The lowest estimate among the four methods. Replacing every missing value with the same constant artificially compresses the variance of the dependent variable, attenuating all slope coefficients toward zero. Mean imputation is generally a poor strategy and should be avoided.

Predicted value imputation (0.0624): Identical to the complete-cases estimate, as expected—the imputed values are exactly the fitted values from that regression, so OLS recovers the same coefficients. The standard errors are artificially small because residual variance is understated.

Multiple imputation (0.0590): The closest estimate to the true value among the four methods. By adding stochastic noise to imputed values, MI restores variance in the dependent variable and produces more credible standard errors via Rubin’s rules. Despite the MNAR nature of the data, MI performs best in practice and is the preferred approach.

Overall, the results illustrate the classic attenuation bias from poorly handling missing data. Mean imputation is clearly the worst strategy; multiple imputation is the most reliable.

Question 8: Project Progress

For my final project, I am using **Compustat firm financials data**, which contains annual accounting and financial information for publicly traded U.S. firms, including variables such as total assets, revenues, capital expenditures, leverage, and profitability measures across a long panel of firms and years.

My primary modeling approach will be **Ordinary Least Squares (OLS)** regression to examine the relationship between key firm financial characteristics and outcomes of interest such as investment or firm growth. Depending on the structure of the data, I may also explore panel fixed effects specifications to control for time-invariant firm heterogeneity.